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Extended-reality visualization of endovascular navigation

Abstract

Techniques are disclosed for extended-reality (XR) visualization of a flexible medical tool. In one example, sensor(s), disposed near a distal portion of a flexible shaft of a medical tool configured for insertion into a patient, sense values indicative of a pose of the medical tool with respect to an EM field. A computing device adjusts, based on the sensed values, a relative alignment between the medical tool, an internal anatomy of the patient, and an exterior of the body of the patient. The computing device generates, based on the computed relative alignment between the medical tool, the internal anatomy, and the exterior of the body, XR content comprising a visual representation of the medical tool within the patients internal anatomy, and outputs the XR content for display to a user so as to simulate a current pose of the medical tool relative to the body of the patient.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6368285	12/2001	Osadchy et al.	N/A	N/A
6725079	12/2003	Zuk et al.	N/A	N/A
7781724	12/2009	Childers et al.	N/A	N/A
8050523	12/2010	Younge et al.	N/A	N/A
8287481	12/2011	Kahn et al.	N/A	N/A
8298147	12/2011	Huennekens et al.	N/A	N/A
8346344	12/2012	Pfister et al.	N/A	N/A
8347738	12/2012	Tung et al.	N/A	N/A
8628491	12/2013	Kahn et al.	N/A	N/A
8852130	12/2013	Govari	N/A	N/A
9072596	12/2014	Kahn et al.	N/A	N/A
9636040	12/2016	Duindam et al.	N/A	N/A
9785246	12/2016	Isaacs et al.	N/A	N/A
9788909	12/2016	Larkin et al.	N/A	N/A
9795319	12/2016	Lavallee et al.	N/A	N/A
9901406	12/2017	State et al.	N/A	N/A
10016148	12/2017	Li	N/A	A61B 34/20
10092215	12/2017	Salamini et al.	N/A	N/A
10123755	12/2017	Walker	N/A	A61B 8/12
10139920	12/2017	Isaacs et al.	N/A	N/A
10143526	12/2017	Walker	N/A	A61B 34/30
10188467	12/2018	Razzaque et al.	N/A	N/A
10194131	12/2018	Casas	N/A	N/A

10271909	12/2018	Guthart et al.	N/A	N/A
10314559	12/2018	Razzaque et al.	N/A	N/A
10405927	12/2018	Lang	N/A	N/A
10444855	12/2018	Isaacs et al.	N/A	N/A
10478255	12/2018	West et al.	N/A	N/A
10481704	12/2018	Parazynski	N/A	N/A
10788791	12/2019	Gellman et al.	N/A	N/A
2002/0198588	12/2001	Armstrong et al.	N/A	N/A
2005/0020963	12/2004	Gabal	N/A	N/A
2008/0013809	12/2007	Zhu et al.	N/A	N/A
2008/0091101	12/2007	Velusamy et al.	N/A	N/A
2010/0249506	12/2009	Prisco	N/A	N/A
2012/0220879	12/2011	Fandrey et al.	N/A	N/A
2014/0189508	12/2013	Granchi et al.	N/A	N/A
2014/0282008	12/2013	Verard et al.	N/A	N/A
2016/0324580	12/2015	Esterberg	N/A	N/A
2017/0014194	12/2016	Duindam et al.	N/A	N/A
2017/0119481	12/2016	Romo	N/A	A61B 1/00096
2017/0367771	12/2016	Tako et al.	N/A	N/A
2018/0116732	12/2017	Lin et al.	N/A	N/A
2018/0185100	12/2017	Weinstein et al.	N/A	N/A
2018/0200018	12/2017	Silva et al.	N/A	N/A
2018/0214138	12/2017	Prisco	N/A	A61B 17/3478
2018/0279852	12/2017	Rafii-Tari	N/A	G16H 40/63
2018/0303563	12/2017	West et al.	N/A	N/A
2019/0155439	12/2018	Mukherjee	N/A	A63F 13/214
2019/0246088	12/2018	Casas	N/A	N/A
2019/0328458	12/2018	Shmayahu	N/A	A61B 5/064
2020/0275988	12/2019	Johnson et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2013116140	12/2012	WO	N/A

OTHER PUBLICATIONS

Battisti et al., “Fiber Bragg gratings-based sensing for real-time needle tracking during MR-guided brachytherapy”, Medical Physics, vol. 43, No. 10, Oct. 2016, 10 pp., doi: 10.1118/1.4961743. cited by applicant

Clarke et al., “Virtual Reality Imaging with Real-Time Ultrasound Guidance for Facet Joint Injection: A Proof of Concept”, International Anesthesia Research Society, vol. 110, No. 5, Dec. 29, 2009, pp. 1461-1463. cited by applicant

Heinrich et al., “HoloInjection: augmented reality support for CT-guided spinal needle injections”, Healthcare Technology Letters, vol. 6, No. 6, Nov. 26, 2019, pp. 165-171, doi: 10.1049/htl.2019.0062. cited by applicant

International Preliminary Report on Patentability from International Application No. PCT/US2021/070787 dated Jan. 12, 2023, 9 pp. cited by applicant

International Search Report and Written Opinion of International Application No. PCT/US2021/070787 dated Oct. 22, 2022, 9 pp. cited by applicant

Kamarudin et al., “Augmented Reality, Virtual Reality, and Mixed Reality in Medical Education: A Comparative Web of Science Scoping Review”, Preprints, Apr. 29, 2019, 10 pp.,

doi:10.20944/preprints201904.0323.v1. cited by applicant

Liu et al., "An augmented reality system for image guidance of transcatheter procedures for structural heart disease", PLoS One, vol. 14, No. 7, e0219174, PubMed, Jul. 1, 2019, 16 pp., doi: 10.1371/journal.pone.0219174. eCollection 2019. cited by applicant

Solbiati et al., "Augmented reality for interventional oncology: proof-of-concept study of a novel high-end guidance system platform", European Radiology Experimental, vol. 2, No. 18, Jul. 31, 2018, 9 pp., <https://doi.org/10.1186/s41747-018-0054-5>. cited by applicant

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Background/Summary

(1) This application claims the benefit of U.S. Provisional Patent Application No. 63/045,542, entitled "AUGMENTED REALITY VISUALIZATION OF ENDOVASCULAR NAVIGATION," and filed Jun. 29, 2020, and also claims the benefit of U.S. Provisional Patent Application No. 63/200,007, entitled "EXTENDED-REALITY VISUALIZATION OF ENDOVASCULAR NAVIGATION," and filed Feb. 9, 2021. The entire contents of U.S. Provisional Patent Application Nos. 63/045,542 and 63/200,007 are incorporated herein by reference.

TECHNICAL FIELD

(1) This disclosure generally relates to systems for visualizing procedures involving implantable medical articles.

BACKGROUND

(2) Clinicians use medical imaging techniques to visualize the interior of a body of a patient for clinical analysis and medical treatment. Medical imaging may be used to implement image-guided clinical procedures, in which a clinician uses preoperative or intraoperative images of the patient to guide the positioning and use of medical instruments. Systems for performing image-guided clinical procedures may use cameras, ultrasonic imaging, electromagnetic imaging, X-ray imaging, or other types of imaging to capture and relay the position of medical tools with respect to the anatomy of the patient. Typically, a clinician performs such image-guided clinical procedures in real-time. For example, fluoroscopy is an example medical imaging technique that uses X-rays to obtain real-time moving images of the interior of the body of the patient. A clinician may use images obtained by fluoroscopy to visualize, for example, a medical tool implanted within the patient and guide navigation of the medical tool to a target site within the patient's anatomy.

SUMMARY

(3) Techniques are disclosed for extended-reality (XR) visualization of a pose (e.g., a relative position and orientation) of a flexible medical tool configured for insertion into a patient. The techniques set forth herein may enable XR visualization of a position of medical tool within an interior anatomy of a patient and with respect to an exterior anatomy of the patient, such as during a clinical procedure. In some examples, the medical tool is a catheter, a needle, a sheath, a guidewire, or a transjugular intrahepatic portosystemic shunt (TIPS) set.

(4) In one example of the techniques of the disclosure, one or more electromagnetic (EM) sensor(s) of an EM sensor system are disposed near a distal portion of a flexible shaft of the medical tool. The EM sensors generate signals indicative of a magnitude or strength of an EM field generated by an EM source of the sensor system. Processing circuitry receives the signals indicative of the magnitude of the EM field and determines a pose of the one or more EM sensors, and therefore a pose of the distal portion of the flexible shaft of the medical tool. The processing circuitry adjusts a

three-dimensional (3-D) model representation of the medical tool based on the determined pose of the one or more EM sensors.

(5) The processor receives first image data depicting at least a portion of an internal anatomy of the body of the patient and an internal imaging target. The processor further receives second image data depicting an external imaging target and/or a portion of an exterior of the body of the patient. The processor correlates the signals indicative of the magnitude or strength of the EM field, the first image data, and the second image data to one another to determine a relative alignment between the pose of the one or more EM sensors, a pose of the external imaging target, and a pose of the internal imaging target. In some examples, a pose of the EM sensor system, the internal imaging target, and the external imaging target are positioned at predetermined distances from one another such that the processing circuitry may map the signals indicative of the magnitude of the EM field, the internal imaging target depicted in the first image data, and the external imaging target depicted in the second image data to map the pose of the one or more EM sensors, the pose of the external imaging target, and the pose of the internal imaging target to one another. The computing device generates, based on the determined relative alignment, XR content comprising a visual representation of the medical tool within the at least a portion of the internal anatomy of the body. The processor output the XR content to an XR device, such as a head-mounted display (HMD), which displays the XR content to a user, such as overlaid onto a view of the patient's body so as to simulate a real-time position and orientation of the medical tool.

(6) The techniques of the disclosure may provide specific improvements to the computer-related field of XR visualization systems that have practical applications. For example, the techniques described herein provide for surgical guidance systems that may be significantly more accurate and/or precise than previous systems. More specifically, the techniques described herein include computer-implemented methods for defining and precisely aligning multiple distinct coordinate systems based on highly accurate sensor data, thereby enabling graphical visualizations representative of a patient's internal anatomy and a medical tool aligned according to their real-world positions and orientations with respect to each other in real-time.

(7) In one example, this disclosure describes a method including: receiving, by processing circuitry and from one or more electromagnetic (EM) sensors of an EM sensor system, signals indicative of a magnitude of an EM field generated by an EM source of the EM sensor system, wherein the one or more EM sensors are disposed at a distal portion of a flexible shaft of a medical tool configured for insertion into a body of a patient; determining, by the processing circuitry and based on the signals indicative of the magnitude of the EM field, a pose of the one or more EM sensors; adjusting, by the processing circuitry and based on the determined pose of the one or more EM sensors, a three-dimensional (3-D) model representation of the medical tool; receiving, by the processing circuitry, first image data depicting at least a portion of an internal anatomy of the body of the patient and an internal imaging target; receiving, by the processing circuitry, second image data depicting an external imaging target; determining, by the processing circuitry and based on the signals indicative of the magnitude of the EM field, the first image data, and the second image data, a relative alignment between the pose of the one or more EM sensors, a pose of the external imaging target, and a pose of the internal imaging target; generating, by the processing circuitry and based on the determined relative alignment, extended reality (XR) content comprising a visual representation of the medical tool within the at least a portion of the internal anatomy of the body; and outputting, by the processing circuitry, the XR content for display on an XR device.

(8) In another example, this disclosure describes processing circuitry of a computing device configured to receive, from one or more electromagnetic (EM) sensors of an EM sensor system, signals indicative of a magnitude of an EM field generated by an EM source of the EM sensor system, wherein the one or more EM sensors are disposed at a distal portion of a flexible shaft of a medical tool configured for insertion into a body of a patient; determine, based on the signals indicative of the magnitude of the EM field, a pose of the one or more EM sensors; adjust, based on

the determined pose of the one or more EM sensors, a three-dimensional (3-D) model representation of the medical tool; receive first image data depicting at least a portion of an internal anatomy of the body of the patient and an internal imaging target; receive second image data depicting an external imaging target; determine, based on the signals indicative of the magnitude of the EM field, the first image data, and the second image data, a relative alignment between the pose of the one or more EM sensors, a pose of the external imaging target, and a pose of the internal imaging target; generate, based on the determined relative alignment, extended reality (XR) content comprising a visual representation of the medical tool within the at least a portion of the internal anatomy of the body; and output the XR content for display on an XR device.

(9) In another example, this disclosure describes a system comprising: a medical tool configured for insertion into a body of a patient; an electromagnetic (EM) sensor system comprising: an EM source configured to generate an EM field; and one or more EM sensors configured to generate signals indicative of a magnitude of the EM field, wherein the one or more EM sensors are disposed at a distal portion of a flexible shaft of the medical tool; processing circuitry configured to: receive, from the one or more EM sensors, the signals indicative of the magnitude of the EM field; determine, based on the signals indicative of the magnitude of the EM field, a pose of the one or more EM sensors; adjust, based on the determined pose of the one or more EM sensors, a three-dimensional (3-D) model representation of the medical tool; receive first image data depicting at least a portion of an internal anatomy of the body of the patient and an internal imaging target; receive second image data depicting an external imaging target; determine, based on the signals indicative of the magnitude of the EM field, the first image data, and the second image data, a relative alignment between the pose of the one or more EM sensors, a pose of the external imaging target, and a pose of the internal imaging target; generate, based on the determined relative alignment, extended reality (XR) content comprising a visual representation of the medical tool within the at least a portion of the internal anatomy of the body; and the XR device, where in the XR device is configured to display the XR content to a user.

(10) The details of one or more examples of the techniques of this disclosure are set forth in the accompanying drawings and the description below. Other features, objects, and advantages of the techniques will be apparent from the description and drawings, and from the claims.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a block diagram illustrating an example system for generating extended-reality (XR) content for visualizing a flexible medical tool.
- (2) FIG. 2 is a block diagram illustrating the flexible medical tool of FIG. 1 in further detail.
- (3) FIGS. 3A-3D are conceptual diagrams illustrating an example technique for generating an XR rendering of the flexible medical tool of FIG. 1.
- (4) FIGS. 4A-4D are images depicting example XR renderings of the flexible medical tool of FIG. 1.
- (5) FIG. 5 is a block diagram illustrating another example system for generating XR content for visualizing a flexible medical tool.
- (6) FIGS. 6A and 6B are conceptual diagrams illustrating another example system for generating XR content for visualizing a flexible medical tool, in accordance with techniques of this disclosure.
- (7) FIGS. 7A-7C is a conceptual diagram illustrating examples of multi-functional fiducial markers.
- (8) FIGS. 8A and 8B are flowcharts illustrating example techniques for determining a relative alignment for an array of fiducial markers.
- (9) FIGS. 9A-9D are conceptual diagrams illustrating the technique of FIG. 8A.

(10) FIG. 10 is a flowchart illustrating an example technique for generating XR content for a medical procedure.

(11) Like reference characters refer to like elements throughout the figures and description.

DETAILED DESCRIPTION

(12) Clinicians typically review a pre-operative computerized tomography (CT) scan of a patient to build a conceptual model of the patient's anatomy prior to performing the operation. The clinician may use other imaging modalities, such as ultrasound, during the operation (e.g., “intraoperatively”) to direct a plane of view (e.g., in examples in which the ultrasound imagery corresponds to a plane) within which a medical tool, such as a needle, sheath, or catheter, is traveling. Current imaging modalities, such as fluoroscopy and ultrasound, only provide operators with a two-dimensional (2-D) view of the field. These techniques have varying degrees of success, and depend on the clinician's ability to mentally reconstruct a 3-D model of the patient's internal anatomy, e.g., based on CT-scan imagery.

(13) “Artificial” reality or “extended” reality (XR) systems are becoming increasingly ubiquitous with applications in many fields such as computer gaming, health and safety, industry (e.g., manufacturing, agriculture, etc.), and education. In general, XR is a form of reality that has been adjusted in some manner before presentation to a user. XR may include, e.g., a virtual reality (VR), an augmented reality (AR), a mixed reality (MR), a hybrid reality, or any combination and/or derivatives thereof. Typical XR systems include one or more devices for rendering and displaying content to users. As one example, an XR system may incorporate a head-mounted display (HMD) worn by a user and configured to output XR content to the user. In particular, the XR system typically computes a current pose (e.g., relative position and orientation) within a frame of reference, such as relative to (e.g., from a point of view of) the HMD, and selectively renders XR content for display to the user based on the current pose. The XR content may include content that is entirely artificial, or content that includes some artificially generated content combined with “captured” content, such as real-world video and/or images.

(14) Techniques are described herein for generating XR content for visualizing a real-time position and orientation of a medical tool, for example, while the medical tool is inserted within a patient during a clinical procedure. In one example, the medical tool includes a Transjugular Intrahepatic Portosystemic Shunt (TIPS) set. The medical tool may incorporate a modified sheath (which may be referred to herein as a “catheter”) and/or a puncture needle, outfitted with one or more sensors along a flexible shaft. In one implementation, the sheath or needle is outfitted with (e.g., is mechanically coupled to) sensors. In other implementations, the sensors may be printed directly onto the sheath and/or needle. The sensors along the shaft provide position information to imaging-overlay software executed by a computing device, which uses the position information to alter an XR image overlay of the medical tool so as to correspond to the actual physical position and/or orientation of the needle and/or sheath. The medical tool further incorporates a fiducial marker (also referred to herein as an “imaging target”), which may be positioned at a base of the medical tool. The function of the imaging target is to provide a point of reference to the XR display device, such as a headset or additional cameras located in the periphery of the surgical field, with a fiducial or reference point from which to position the XR-image overlay. Although described herein with respect to TIPS procedures, the techniques of this disclosure may be applied to any suitable medical procedure including, but not limited to, a prostate artery embolization procedure, a genicular artery embolization procedure, or the like.

(15) In some examples, the techniques described herein include using 3-D information obtained from preoperative CT scans in order to allow a clinician to better understand spatial relationships of target anatomy of a patient. The techniques of the disclosure may have many other commercial applications, which may include percutaneous endovascular targeting, XR-guided endovascular navigation, or percutaneous biopsy tracking of a flexible needle.

(16) FIG. 1 is a block diagram illustrating example system 100 for generating XR content 112 for

visualizing flexible medical tool **102**. System **100** includes medical tool **102**, imaging device **120**, computing device **106**, and XR head-mounted display (HMD) **108**.

(17) Medical tool **102** is typically a flexible medical instrument capable of insertion or implantation within a patient. In some examples, medical tool **102** comprises a catheter, a needle, or a sheath. Medical tool **102** includes flexible shaft **114** and base **116**. A plurality of sensors **104** are disposed longitudinally along flexible shaft **114**. Each of the plurality of sensors **104** is configured to sense a plurality of values. Each value of the plurality of values is indicative of a change in a position of a respective portion of shaft **114** at which a corresponding sensor **104** is disposed, wherein the change in position is measured from a “calibration” position of the respective portion of shaft **114**.

(18) In some examples, base **116** of medical tool **102** is a base of a modified puncture needle. Base **116** is configured to receive signals from sensors **104** disposed on shaft **114**, and includes communication circuitry for communicating sensor signals from sensors **104** to computing device **106** and/or HMD **108**. Furthermore, base **116** may include, or act as, a fiducial marker **118** from which computing device **106** overlays a digital representation **152** of medical tool **102**, as described in more detail below.

(19) Imaging device **120** includes, as examples, a camera or a video-capture device. Imaging device **120** generates one or more images of medical tool **102** within the 3-D environment. In some examples, imaging device **120** generates images that depict or otherwise include fiducial marker **118** affixed to base **116** of medical tool **102**.

(20) Computing device **106** includes processing circuitry and memory. In some examples, the processing circuitry of computing device **106** includes one or more processors, such as one or more microprocessors, digital signal processors (DSPs), application-specific integrated circuits (ASICs), field-programmable gate arrays (FPGAs), or any other equivalent integrated or discrete logic circuitry, as well as any combinations of such components. In some examples, the memory of computing device **106** includes random-access memory (RAM), read-only memory (ROM), programmable read-only memory (PROM), erasable programmable read-only memory (EPROM), electronically erasable programmable read-only memory (EEPROM), or flash memory, and includes executable instructions for causing the one or more processors to perform the actions attributed to them, as described herein. Further, this memory may be implemented entirely in hardware, software, or a combination thereof.

(21) Medical tool **102** transmits, to computing device **106**, the values sensed by the plurality of sensors **104**. Computing device **106** receives the plurality of values sensed by the plurality of sensors **104** and adjusts, based on the values sensed by the plurality of sensors **104**, a 3-D model representation **152** of medical tool **102**. For example, computing device **106** may rotate, translate, transform, stretch, warp, or bend a 3-D model representation **152** of medical tool **102** in order to approximate or match a real-world shape, configuration, or position of medical tool **102** as sensed by sensors **104**. For example, sensors **104** may detect a flexing or bending of medical tool **102** away from a “calibration” configuration of medical tool **102**.

(22) Computing device **106** uses the images of medical tool **102** from imaging device **120** in order to compute a pose of medical tool **102** within the 3-D environment. For example, computing device **106** uses images depicting fiducial marker **118** positioned on medical tool **102** as a fiducial reference point to localize a position and/or orientation of medical tool **102** within a 3-D, real-world environment with respect to the patient. Computing device **106** uses the adjusted 3-D model representation of medical tool **102** and the images of medical tool **102** from imaging device **120** to generate XR content **112** including a visual representation **152** of medical tool **102** within the 3-D environment. In other words, computing device **106** may execute software code that interprets the information from sensors **104** to create an altered digital structure representing medical tool **102**. Computing device **106** may alter a digital replica or model of medical tool **102** to conform to the real-world physical structure of medical tool **102**, such as based on the information provided by sensors **104** located along shaft **114**. For example, XR content **112** may include a visual

representation **152** of medical tool **102** (e.g., the adjusted 3-D model representation of medical tool **102**) overlaid upon an image of the real-world environment, so as to represent a real-world position and shape of medical tool **102** within a body of the patient.

(23) Computing device **106** outputs the XR content to a display device for display to user **110**. As depicted in the example of FIG. **1**, the display device includes head-mounted display (HMD) **108**. In other examples, the display may be a computer monitor, mobile device, television, or other type of stationary, screen-mounted display.

(24) FIG. **2** is a block diagram illustrating flexible medical tool **102** of FIG. **1** in further detail. In some examples, medical tool **102** includes a catheter. In some examples, medical tool **102** includes a needle and/or a guidewire. As depicted in FIG. **2**, medical tool **102** includes flexible shaft **114** and base **118**.

(25) Sensors **104A-104E** (collectively referred to hereinafter as “sensors **104**”) are disposed longitudinally along shaft **114**. In some examples, sensors **104** are disposed longitudinally along a centerline axis of flexible shaft **114** (e.g., such as within a core of shaft **114**). In some examples, sensors **104** are disposed longitudinally along a first axis parallel to a centerline axis of the flexible shaft (e.g., such as along an exterior sheath of shaft **114**). In some examples, sensors **104** are disposed longitudinally along at least a first axis parallel to a centerline axis of the flexible shaft and a second axis parallel to the centerline axis of the flexible shaft and offset from the first axis. For example, two of sensors **104** may be positioned along each portion of shaft **114** and spaced 90 degrees apart so as to measure multiple axes of motion of shaft **114**. As another example, three sensors **104** may be positioned along each portion of shaft **114** and spaced 60 degrees apart so as to measure multiple axes of motion of shaft **114**. In some examples, a 3-D printer is used to directly print sensors **104** upon shaft **114** of medical tool **102**.

(26) Each of sensors **104** senses a flexing or bending of a respective portion of shaft **114** upon which the sensor **104** is disposed. In some examples, sensors **104** include flex sensors capable of sensing a flexing or bending of shaft **114**. In some examples, sensors **104** include strain gauges. Sensors **104** may be analog sensors or digital sensors. For example, each of sensors **104** senses a value indicative of a calibration position of the respective portion of the shaft **114** at which the sensor is disposed. Further, each of sensors **104** senses a value indicative of a change in a relative position of a respective portion of shaft **114** at which a corresponding sensor **104** is disposed, as measured from a calibration position of the respective portion of shaft **114**. Sensors **104** output values indicative of the change in the position of the respective portion of shaft **114** at which each sensor **104** is disposed, as measured from the calibration position of the respective portion of shaft **114**. In some examples, the sensed value is indicative of a displacement of the position of the respective portion of shaft **114** at which the sensor **104** is disposed, as measured from the calibration position of the portion of shaft **114**. In some examples, the sensed value is indicative of an angular displacement of the position of the respective portion of shaft **114** at which the sensor **104** is disposed, as measured from the calibration position of the respective portion of shaft **114**.

(27) Base **116** includes communication circuitry **204** for transmitting, to computing device **106** (FIG. **1**), the values sensed by the plurality of sensors **104**. In some examples, communication circuitry **204** communicates with computing device **106** via a wired or wireless connection. Base **116** further includes fiducial marker **118**, which computing device **106** may use as a fiducial reference point to localize a position and/or orientation (e.g., a “pose”) of medical tool **102** within a 3-D, real-world environment with respect to the patient.

(28) FIGS. **3A-3D** are conceptual diagrams illustrating an example technique for generating an XR rendering **152** of flexible medical tool **102**. As depicted in FIGS. **3A** and **3B**, computing device **106** (FIG. **1**) overlays a 3-D model representation **152** of medical tool **102** over an image of medical tool **102**. In some examples, computing device **106** uses a fiducial marker or imaging target **118** (which may or may not be fixedly positioned on medical tool **102**) in order to localize a position and/or orientation of medical tool **102**, so as to correctly position 3-D model **152** overtop of

medical tool **102** within the image. As shown in FIGS. 3C and 3D, computing device **106** adjusts a shape of 3-D model representation **152** based on the values sensed by sensors **104** so as to mold 3-D model **152** to approximate or conform to an actual curve or shape of the physical medical tool **102**.

(29) FIGS. 4A-4D are diagrams depicting example XR renderings of flexible medical tool **102** of FIG. 1. For example, FIGS. 4A-4D depict real-world images of medical tool **102** of FIG. 1, and rendered representations **404** of a relative position of each sensor **104**, shown overlaid upon the real-world image. As medical tool **102** flexes and bends in FIG. 4C, the rendered representation of each sensor **104** similarly moves in order to depict the real-world position of the corresponding portion of medical tool **102** (FIG. 4D).

(30) FIG. 5 is a block diagram illustrating another example system for generating XR content for visualizing a flexible medical tool **102**. In the example of FIG. 5, medical tool **102** includes a plurality of sensors **104** disposed longitudinally along a flexible shaft of medical tool **102**. Sensors **104** are configured to sense a plurality of values. Each value of the plurality of sensed values is indicative of a change in a position of a respective portion of the shaft of medical device **102**, at which a corresponding sensor **104** of the plurality of sensors **104** is disposed, wherein the change in position is measured from a calibration position of the portion of the shaft of medical device **102**.

(31) Processing circuitry **107** of computing device **106** receives the values sensed by the plurality of sensors and adjusts, based on the sensed plurality of values, a three-dimensional (3-D) model representation of medical tool **102**. Computing device **106** further receives, from camera **120** (which may be composed of a single unit or a plurality of units providing visualization of the device **102** from a plurality of perspectives), one or more images of medical tool **102** within the 3-D environment. Computing device **106** then executes programming **109** configured to compute, based on the one or more images, a pose of medical tool **102** within the 3-D environment.

Computing device **106** generates, based on the adjusted 3-D model representation of medical tool **102** and the computed pose of medical tool **102** within the 3-D environment, a visual representation **152** of medical tool **102** within the 3-D environment. Computing device **106** generates XR content comprising the visual representation of medical tool **102** and outputs the XR content to a display device, such as HMD **108** or a stationary screen, for display to user **110**.

(32) FIGS. 6A and 6B are conceptual diagrams illustrating another example XR system **600** for generating XR content for visualizing a flexible medical tool, in accordance with techniques of this disclosure. System **600** includes medical tool **602**, computing device **606**, EM sensor system **650**, external imaging device **608**, internal imaging device **610**, and head-mounted display (HMD) **108**.

(33) XR system **600** (e.g., via processing circuitry of HMD **108** or another computing device **606**) is configured to use three sets of information to generate XR content **612**: (1) electromagnetic sensor data captured by EM sensor system **650**, indicative of a pose of medical tool **602**; (2) “external” image data depicting an exterior of a patient's body **620**, as captured by external imaging device **608**; and (3) “internal” image data depicting an interior anatomy of the patient's body **620**, as captured by internal imaging device **610**. XR content **612** includes at least a visual representation **652** of medical tool **602** positioned “accurately” within (e.g., reflecting a real-world and real-time position of medical tool **602** relative to) a visual representation **654** of at least a portion of the internal anatomy of a patient's body **620**.

(34) Based on these three sets of data, system **600** is configured to display XR content **612** so as to generate a “transparent view” through the exterior of the patient's body **620** that graphically depicts a real-time simulation of the actual position and orientation of medical tool **602** within the patient's internal anatomy. In examples in which HMD **108** includes a “transparent” visor or lens with XR display functionality, system **600** is configured to display XR content **612** relative to the actual exterior of the patient's body **620**, as viewed through the transparent lens. In other examples in which HMD **108** includes an “opaque” display screen (e.g., a virtual-reality headset or goggles), system **600** is configured to generate composite imagery by overlaying XR content **612** onto 2-D

imagery depicting the exterior of the patient's body **620**, and display the composite imagery on the display screen of HMD **108**.

(35) Medical tool **602** is an example of medical tool **102** of FIG. **1**, and is typically a flexible medical instrument capable of insertion or implantation within a patient **620**. In some examples, medical tool **602** includes a catheter, a needle, a guidewire, or a sheath. Medical tool **602** includes a flexible, elongated shaft **614** and a base **616** at a proximal end of the shaft **614**. In some examples, base **616** of medical tool **602** is a base of a modified puncture needle.

(36) System **600** includes one or more “external” imaging targets **618** (also referred to herein as “markers” or “fiducial targets”) that are visible to (e.g., detectable by) external imaging device **608**, and one or more “internal” imaging targets **628** that are visible to (e.g., detectable by) internal imaging device **610**. For instance, as shown in FIG. **6A**, external imaging target **618** includes a visually distinct marker that is detectable within external imagery captured by external imaging device **608**, which may be an example of imaging device **120** of FIG. **1**. For instance, external imaging device **608** may include a forward-facing camera of HMD **108**.

(37) Similarly, internal imaging target **628** includes a visually distinct marker that is detectable within internal imagery captured by internal imaging device **610**, which may also be an example of imaging device **120** of FIG. **1**. For instance, internal imaging device **610** may include an X-ray, MRI machine, CT scanner, or other device configured to generate internal imagery depicting, describing, or otherwise indicative of at least a portion of the internal anatomy of patient **620**. In such examples, internal imaging target **628** may be positioned next to (e.g., external to) or within (e.g., internal to) patient **620** as internal imaging device **610** generates the internal imagery. In either example, internal imaging target **628** should be “fixed” with respect to at least one other structure to provide a reliable point of reference. For instance, internal imaging target **628** may be fixed to a surface of a patient bed while internal imaging device **610** captures the internal imagery. In other examples, internal imaging target **628** may include, or may be rigidly coupled to, a “consistent” portion of the patient's internal or external anatomy.

(38) External imaging target(s) **618** and internal imaging target(s) **628** provide reference points within the three datasets used by system **600** to enable **600** to “align” or orient the various datasets relative to one another. For instance, in the example depicted in FIG. **6A** (but not in all examples), external imaging target **618** comprises a visual marker that is rigidly coupled to (or defined by) the base **616** of medical tool **602**. In such examples, external imaging target **618** provides a “dual” common reference point for alignment of the external imaging data (e.g., based on the position of external imaging target **618** relative to the body of patient **620** when depicted in the external imaging data) with the EM sensor data (e.g., based on the fixed and known position of external imaging target relative to medical tool **602**).

(39) In some examples, a single coherent physical object may be configured to provide the dual-reference-point functionality of both external imaging target **618** and internal imaging target **628**, e.g., that is visible to both external imaging device **608** and internal imaging device **610** for alignment of those two datasets. As one non-limiting example, external imaging target **618** may be formed from particular materials, such that it is both visibly opaque to the visible light spectrum when depicted in camera imagery (e.g., ink, or any other optically visible material), as well as radiopaque, e.g., within x-ray imagery, MM imagery, CT scan imagery, or any other type of internal imaging data (e.g., metal, lead, etc.).

(40) In some examples, external imaging target **618** may either be rigidly coupled to medical tool **602** (as described above), and/or may additionally include one or more EM sensors **604**, so as to provide a dual reference point for alignment of the external imaging data (depicting at least a portion of the exterior of the patient's body **620**) and for EM data of EM system **650** (describing the relative pose of medical tool **602** within the body of patient **620**).

(41) Similarly, in some examples, internal imaging target **628** may either be rigidly coupled to medical tool **602** (as described above), and/or may include one or more EM sensors **604**, so as to

provide a common reference point for both the internal imaging data (describing the interior anatomy of the patient's body **620**) from internal imaging device **610**, and for the EM data of EM system **650** (describing medical tool **602** within the interior anatomy of the patient's body **620**).

(42) In some examples in accordance with this disclosure, and as detailed further below with respect to FIG. 7, a single coherent object may include suitable physical and/or chemical properties such that the single object constitutes a “treble” common reference point, providing the functionality of all three of (1) an external imaging target **618** with respect to at least a portion of the exterior anatomy of patient **620**; (2) an internal imaging target **628** with respect to the interior anatomy of patient **620**; and (3) an EM reference point (with respect to medical tool **602** within patient **620**). In such examples, the single coherent object provides a common location and orientation in space on which to base an alignment of corresponding coordinate systems for all three sets of data of system **100**, e.g., to facilitate subsequent alignment by system **600** of (1) a virtual model **652** of medical tool **602** relative to a virtual model **654** of the internal anatomy of the body of patient **620**; and (2) the virtual model **654** of the internal anatomy of the body of patient **620** relative to the exterior of the body of patient **620**, either as viewed through a transparent display screen of HMD **108**, or as depicted in 2-D imagery to be displayed on an opaque display screen of HMD **108**.

(43) Medical tool **602** includes electromagnetic (EM) sensor system **650**, which in turn includes EM source **622** and at least one EM sensor **604**. EM source **622** generates an EM field **624** (FIG. 6B). In some examples, but not all examples, EM source **622** may include a plurality of EM antennae, each EM antenna configured to emit an EM field defining a unique EM field pattern to be sensed (and distinguished) by EM sensor **604**.

(44) EM sensor **604** senses a strength (e.g., magnitude, phase, frequency, or the like) of the EM field at the location at which EM sensor **604** is positioned. EM sensor system **650** further includes communication circuitry (not shown) that receives (e.g., via a wired or wireless connection) signals generated by EM sensor(s) **604**, and communicates the EM sensor signals to computing device **606** and/or HMD **108**. Based on the sensed magnitude of the EM field at the location at which EM sensor **604** is positioned, computing device **606** may determine a distance and relative orientation between EM sensor **604** and EM source **622**.

(45) As depicted in the examples of FIGS. 6A and 6B, EM sensor **604** is located at a distal portion (e.g., at or near the tip) of flexible shaft **614** of medical tool **602**. In the example of FIG. 6A, EM source **622** is located at base **616** of medical tool **602**. Therefore, by determining a distance between EM sensor **604** and EM source **622**, computing device **606** may determine a distance between the distal portion of flexible shaft **614** and base **616** of medical tool **602**. By comparison, in the example of FIG. 6B, EM source **622** is located within a device that is physically distinct from medical tool **602**.

(46) In some examples, medical tool **602** may include more than one EM sensor **604**. The use of a first EM sensor **604** in different orientation with respect to a second EM sensor **604** may provide additional information that computing device **606** may use to determine a 6-dimensional position of the distal portion of shaft **614** of medical tool **602** with respect to EM field generator **622**. For example, medical tool **602** may include multiple EM sensors **604** clustered at a distal portion of medical tool **602** and/or distributed along flexible shaft **614**. In some examples, medical tool **602** may include EM sensor(s) **604** near the distal tip of the tool, in addition to other types of sensors distributed along flexible shaft **614** (e.g., such as flex or strain sensors **104** of FIG. 1).

(47) In some examples, the communication circuitry of EM sensor system **650** is located within base **616** of medical tool **602**. The communication circuitry may be operatively connected to EM sensors **604**, e.g., via a wired or wireless connection. In other examples, each of EM sensor(s) **604** includes respective local communication circuitry configured to transmit the respective EM sensor signals directly to computing device **606** and/or HMD **108**.

(48) Computing device **606** is an example of computing device **106** of FIG. 1 and includes one or

more processors (e.g., “processing circuitry”) and memory. In some examples, the processing circuitry of computing device **606** includes one or more microprocessors, digital signal processors (DSPs), application-specific integrated circuits (ASICs), field-programmable gate arrays (FPGAs), or any other equivalent integrated or discrete logic circuitry, as well as any combinations of such components. In some examples, the memory of computing device **606** includes random-access memory (RAM), read-only memory (ROM), programmable read-only memory (PROM), erasable programmable read-only memory (EPROM), electronically erasable programmable read-only memory (EEPROM), or flash memory, and includes executable instructions for causing the one or more processors to perform the actions attributed to them, as described herein. Further, this memory may be implemented entirely in hardware, software, or a combination thereof.

(49) External imaging device **608** generates external image data depicting external imaging target **618** and, in some examples (but not all examples), at least a portion of the exterior of the patient's body **620**. The external image data may further include one or more images of at least a portion of medical tool **602** within the 3-D environment (e.g., relative to the body of patient **620**) or a distinct external imaging target **618**. External imaging device **608** includes, as examples, a camera or a video-capture device. In some examples, external imaging device **608** is an example of imaging device **120** of FIG. **1**. External imaging device **608** may additionally include an infrared-based imaging device, e.g., for infrared tracking of one or more groups of retroreflecting markers.

(50) Internal imaging device **610** generates internal image data depicting at least a portion of an internal anatomy of the body of patient **620** and internal imaging target **628**. Internal imaging device **610** includes, as examples, an x-ray fluoroscopy device, a magnetic resonance imaging (MM) device, a computed tomography (CT) scan device, an ultrasonic imaging device, or the like. In some examples, internal imaging device **610** is an example of imaging device **120** of FIG. **1**.

(51) As depicted in the examples of FIGS. **6A** and **6B**, system **600** includes head-mounted display (HMD) **108**. In other examples, system **600** includes another type of display, such as a computer monitor, mobile device, tablet, laptop, television, transparent computer display, projector, a transparent display with a beamsplitter, or other type of screen-mounted display.

(52) In accordance with techniques of this disclosure, EM system **650** transmits, to computing device **606**, the values (e.g., EM sensor signals) sensed by EM sensor(s) **604** indicative of the magnitude of the EM field generated by EM source **622**. Computing device **606** receives the plurality of values sensed by the EM sensor(s) **604** and determines, based on the EM field sensed by EM sensor(s) **604**, a current “pose” or configuration of medical tool **602**. For example, computing device **606** may determine, based on a magnitude of the EM field (as sensed by EM sensor **604** and as produced by EM source **622** located within base **616** of medical tool **602**), a relative position of the distal portion of medical tool **602** with respect to base **616** of medical tool **102**. While computing device **602** may use a single EM sensor **602** to determine the distance of the distal portion of medical tool **602** with respect to base **616**, computing device **602** may use one or more EM sensors **604** to determine both a position and orientation (e.g., a pose) of the distal portion of medical tool **602** with respect to base **616**.

(53) Computing device **606** uses the information from EM sensor(s) **604** to create an altered digital structure representing medical tool **602**. Computing device **606** may alter a digital replica or model of medical tool **602** to conform to the real-world physical structure of medical tool **602**, such as based on the information provided by EM sensor(s) **604** located at least near a distal portion of flexible shaft **614**. In some examples, responsive to determining a current pose of medical tool **602**, computing device **606** generates (or adjusts) a 3-D model representation **652** of medical tool **602**. For example, computing device **606** may translate, rotate, stretch, warp, bend, or otherwise modify a 3-D model representation **652** of medical tool **602** in order to approximate or match a real-world shape of medical tool **602** as determined by computing device **606** based on the signals generated by sensor(s) **604**.

(54) Computing device **606** receives, from internal imaging device **610**, first (internal) image data

depicting an internal anatomy of the body of the patient **620** and internal imaging target **628**. Further, computing device **606** receives, from external imaging device **608**, second (external) image data depicting external imaging target **618** and, in some cases, at least a portion of an exterior of the body of the patient **620**.

(55) Computing device **606** is further configured to determine, based on the signals indicative of the magnitude and direction (and/or phase, frequency, etc.) of the EM field, the first (internal) image data, and the second (external) image data, a relative alignment between the pose of the one or more EM sensors **604**, a pose of the external imaging target **618**, and/or a pose of the internal imaging target **628**. For example, the components of imaging target **618** visible by external imaging device **608** and the components of imaging target **618** visible by internal imaging device **610** overlap one another. Further, EM source **622** of base **616** may overlap imaging target **618**. In other examples, the components of imaging target **618** visible by external imaging device **608**, the components of imaging target **618** visible by internal imaging device **610**, and EM source **622** are positioned a negligible distance from one another or are positioned a predetermined distance from one another. Because imaging target **618** and EM source **622** are a known distance (e.g., zero or a predetermined distance) from one another, computing device **606** may align information captured by EM sensor system **650**, external imaging device **608**, and internal imaging device **610** with respect to one another to generate XR content as described in more detail below.

(56) When determining the relative alignments between the EM sensor system **650**, the internal imaging system (e.g., internal anatomy and internal imaging target **628**), and the external imaging system (e.g., external imaging target **618**), computing system **606** may be configured to determine a first pairwise alignment between any two of these three systems, a second pairwise alignment between any other two of these three systems, and in some examples, combine the two pairwise alignments to determine a common “global” relative alignment or coordinate system.

(57) For instance, in a first example, computing device **606** is configured to determine the global relative alignment by determining a first relative alignment between the pose of the one or more EM sensors **604** (e.g., a pose of EM sensor(s) **604** with respect to an EM coordinate system) and a pose of the internal imaging target **628** (e.g., a pose of target **628** relative to an internal imaging coordinate system). In some such examples, computing system **606** is configured to generate, based on the adjusted 3-D model representation **652** of the medical tool **602** and the first relative alignment between the pose of the one or more EM sensors **604** and the pose of the internal imaging target **628**, the visual representation **652** of the medical tool **602** within the internal anatomy **654** of the patient **620**. For instance, determining the “first” relative alignment may include determining, based on a predefined distance between the EM sensor system and the internal imaging target **628**, the pose of the one or more EM sensors **604** with respect the internal imaging coordinate system, and/or the pose of the internal imaging target **628** with respect the EM coordinate system.

(58) In some examples, computing system **606** is the further configured to determine a “second” relative alignment, such as between the pose of external imaging target **618** and the pose of internal imaging target **628**. For instance, computing device **606** may be configured to determine the “second” relative alignment between the pose of the external imaging target **618** and the pose of the internal imaging target **628** by determining, based on a predefined distance between the internal imaging target **628** and the external imaging target **618**, the pose of the internal imaging target **628** with respect the external coordinate system, or equivalently, the pose of the external imaging target **618** with respect to the internal coordinate system.

(59) Computing device **606** uses the adjusted 3-D model representation **652** of medical tool **602**, the external imagery from external imaging device **608** depicting medical tool **602**, and the internal images of medical tool **602** from internal imaging device **610** to generate XR content **612**, including a visual representation **652** of medical tool **602** with respect to a visual representation **654** of the internal anatomy of patient **620**, and further, with respect to an exterior, 3-D

environment. For example, in cases in which an intended output device includes a transparent display screen (e.g., a transparent visor of HMD **108**), computing device **606** uses images depicting external imaging target **618** positioned on medical tool **602** as a reference point to localize a position and/or orientation (e.g., a pose) of medical tool **602** within a 3-D, real-world environment with respect to patient **620**. Further, computing device **606** uses images depicting internal imaging target **628** positioned on medical tool **602** as a reference point to localize a position and/or orientation of medical tool **602** within the 3-D, internal anatomy image data of patient **620**.

(60) For instance, computing device **606** may use the received signals from EM sensor **604** to determine the pose of medical tool **602** and generate a representation of medical tool **602** in the determined pose, as described above. Further, computing device **606** renders, based on the determined relative alignment, the representation of medical tool **602** positioned within a visual representation of the internal anatomy of the body of patient **620**. Computing device **606** outputs, based on the determined relative alignment, the representation **652** of medical tool **602** and the visual representation **654** of the internal anatomy of the body of patient **620** onto the transparent display screen of HMD **108** at a position that, from the perspective of the user **110**, is “overtop” of the real-time position of the body of the patient **620** in order to simulate a “transparent” view of an accurate position and orientation of tool **602** through the body of the patient **620**. Computing device **606** may further incorporate motion-sensor data from an inertial measurement device (IMD), such as an accelerometer, incorporated within HMD **108** in order to continuously “update” the position and orientation of the XR content **612** on the transparent display screen to “register” or “fix” the XR content in place from the perspective of user **110** during a movement of HMD **108** (e.g., as user **110** moves his or her head).

(61) In other examples in which an intended output device does not include a transparent display screen (e.g., an LCD or other display screen of a computing device, such as a mobile device, VR headset, or the like), the XR content generated by computing device **606** may include a composite image of the visual representation **652** of the medical tool **602** within the internal anatomy **654** overlaid upon a portion of the exterior of the body of the patient **620**, e.g., as depicted within the second (external) image data or other image data depicting a current position and orientation of the patient's body. For instance, computing device **606** uses the received signals from EM sensor **604** to determine the pose of medical tool **602** and generate a representation **652** of medical tool **602** in the determined pose, as described above. Further, computing device **606** renders, based on the determined relative alignment, the representation **652** of medical tool **602** positioned within a visual representation **654** of the internal anatomy of the body of patient **620**. Computing device **606** overlays, based on the determined relative alignment, the representation **652** of medical tool **602** and the visual representation **654** of the internal anatomy of patient **620** over a visual representation of the exterior of the body of patient **620**. Computing device **606** outputs the XR content **612** for display on an XR device (e.g., HMD **108**) for display to user **110**.

(62) As depicted in the example of FIG. **6A**, XR content **612** may include a visual representation **652** of medical tool **602** (e.g., the adjusted 3-D model representation of medical tool **602**) positioned within a visual representation **654** of the internal anatomy of the body of the patient **620**. Further, XR content **612** may depict a visual representation **652** of medical tool **602** and visual representation **654** of the internal anatomy of the body of the patient **620** overlaid upon a visual representation of the real-world environment (e.g., depicting at least a portion of the exterior of the body of patient **620**), so as to present a simulated “transparent view” representing a real-world position and orientation of medical tool **602** within the internal anatomy of the patient **620** (e.g., relative to one or more simulated internal organs of the patient), and relative to the real-world position and orientation of the exterior of the body of patient **620**. In some examples, system **600** may include multiple display devices and/or HMDs **108** that enable multiple users **110** to visualize medical tool **602** within the internal anatomy of patient **620** and within the 3-D real-world environment from the perspective of each user **110**.

(63) FIG. 6B is a conceptual diagram illustrating a portion of XR system **600** of FIG. 6A in further detail. In particular, FIG. 6B illustrates an example of EM sensor system **650** that includes one or more EM sensors **604** coupled to flexible medical tool **602** of system **600**. As shown in the example of FIG. 6B, EM sensor(s) **604** are configured to sense a plurality of signals indicative of a magnitude and direction (and/or phase, frequency, etc.) of EM field **624** generated by EM source **622** of EM sensor system **650**. For example, as detailed further below, one or more EM sensor(s) **604** are configured to sense a magnitude and direction of EM field **624**, from which computing device **606** of FIG. 6 may determine a pose (e.g., relative position and orientation) of EM sensor **604** relative to EM source **622**.

(64) In the example of FIG. 6B, medical tool **602** includes a Transjugular Intrahepatic Portosystemic Shunt (TIPS) set that incorporates a modified sheath or needle outfitted with or more electromagnetic (EM) position sensors **604**. An EM field generator or EM source **622** is located near patient **620** and generates one or more field patterns **624** that are sensed by EM position sensor(s) **604**. Each EM position sensor(s) **604** generates a signal indicative of a position of the EM sensor **604** relative to EM field generator **622** and outputs the signal to an XR system, such as to computing device **606** and/or HMD **108**.

(65) EM field source **622** includes circuitry configured to generate a controlled electromagnetic field **624** having known or controllable parameters (e.g., magnitude, direction, and/or other EM characteristics) configured to be detected by EM sensor(s) **604**. In some cases, such as the example shown in FIG. 6A, EM field source **622** is integrated into a base **616** of medical device **602**. In other examples, such as the example depicted in FIG. 6B, EM field source **622** is depicted as being physically distinct from medical tool **602**. In some examples, EM position sensors **604** determine their position and orientation relative to EM field generator **622**.

(66) In some examples in accordance with this disclosure, computing device **606** is configured to receive, from one or more EM sensors **604** of EM sensor system **650**, signals indicative of a magnitude of an EM field **624** generated by an EM source **622** of the EM sensor system **650**, wherein the one or more EM sensors **604** are disposed at a distal portion of a flexible shaft **614** of medical tool **602** configured for insertion into a body of patient **620**. Computing device **606** and/or another XR system (e.g., HMD **108**) may then determine, based on the signals indicative of the magnitude of the EM field, a pose of the one or more EM sensors **604** and adjust, based on the determined pose of the one or more EM sensors **604**, a three-dimensional (3-D) model representation **652** of the medical tool.

(67) External imaging target **618** and internal imaging target **628** may be attached to EM field generator **622**. In some examples, external imaging target **618** and internal imaging target **628** are the same imaging target. In some examples, either the internal target, the external target, or both targets have respective EM sensors **604**. Imaging targets **618**, **628** provide an imaging marker for an augmented reality (AR) or extended reality (XR) HMD **108**, and associated software, to align data from EM system **650**, internal image data obtained from internal imaging device **610** (of FIG. 6A) and exterior image data obtained from external imaging device **608** (of FIG. 6A) to one another.

(68) When the relative position between the perspective of HMD user **110** and EM position trackers **604** is known, XR software of computing device **606** creates a visual overlay **612** of the operative tools even when occluded by tissue of the body of patient **620**. This allows the user to continuously visualize the position and orientation of the TIPS needle and sheath **602**.

(69) As described above, imaging targets **618**, **628** each include a visually distinct marker that, when depicted within imagery captured by, e.g., external imaging device **608** and/or internal imaging device **610**, computing device **606** can detect the imaging targets **618**, **628** within the imagery, such as by using pattern-recognition algorithms. In some examples, imaging targets **618**, **628** are rigidly coupled to EM field source **622** such that a distance and orientation between imaging targets **618**, **628** and EM field source **622** is known. By extension, computing device **606**

may determine a pose of EM sensor(s) **604** (relative to external imaging target **618**) based on received EM sensor data indicating the pose of EM sensor(s) **604** relative to EM field source **622**. In some cases, system **600** may include a modular mounting system which mechanically registers to both the EM field generator **622** and imaging targets **618**, **628**. This mounting system may be substantially rigid such that, when the EM field generator **622** and/or imaging targets **618**, **628** are installed, their relative poses are known.

(70) In some examples, external imaging target **618** includes an object including three or more elongated arms, each arm having a different length and oriented at a known angle (e.g., about 90 degrees) with respect to the other arms so as to define an unambiguous orientation of external imaging target **618** when observed within external imagery from external imaging device **608** (e.g., a camera). Similarly, each EM sensor **604** may include two or more individual EM sensor units oriented at a known angle (e.g., about 90 degrees) with respect to each other so as to define an unambiguous orientation of the respective EM sensor **604**, as indicated by, and as determinable from, the collected EM sensor signals.

(71) As described above, in some examples, interior imaging target **628** is fixed at a predetermined position relative to a position of the EM sensor system **650** (e.g., rigidly coupled to EM source **622**). Similarly, exterior imaging target **618** may be fixed at a predetermined position relative to the position of the interior imaging target **628**, such that the relative alignments between the various coordinate systems are known or readily determinable. In some examples, positions of any or all of EM source **622**, interior imaging target **628**, and/or exterior imaging target **618** may all be mutually fixed with respect to a rigid portion of the body of patient **620** that is discernable both from external imagery (e.g., visible or determinable through the patient's skin) as well as within the internal imagery. For instance, a rigid portion of the body of patient **620** may include a bony anatomy of the patient, such as the patient's skull or other similar body part. In some such examples, computing system **606** is configured to determine the relative alignment(s) between the datasets based at least in part on known distances of the interior imaging target **628** and the exterior imaging target **618** from the rigid portion of the body of patient **620**. In some such examples, computing system **606** is configured to “register” the bony anatomy of the body of patient **620** to interior imaging target **628**, exterior imaging target **618**, and/or EM sensor(s) **604**. As one particular example of the registration process, EM sensor(s) **604** may be moved or “swiped” along the rigid portion of the patient's body while monitoring the sensed EM signals, such that the location of the rigid portion of the patient's body relative to EM source **622** is known, and by extension (from internal and external imagery of the rigid anatomy portion), the location of EM source **622** relative to the internal coordinate system and/or the external coordinate system.

(72) In some examples, internal imaging target **628** includes an asymmetrical marker (e.g., an object) formed from a visually contrasting material (e.g., a radiopaque material or the like) such that it is discernable within internal imaging data, such as CT scan imagery, MRI imagery, fluoroscopic imagery, ultrasonic imagery, or the like. In some examples, internal imaging target **628** may have a geometric shape that is substantially asymmetrical, such that its orientation is unambiguous when located within the internal imaging data. In some examples, internal imaging target **628** includes an object having at least four points of reference, each point of reference made from a material having a different relative density (measured in “Hounsfield” units) such that the points of reference define an unambiguous six-degrees-of-freedom (6DOF) orientation of internal imaging target **628** in 3-D space. As used herein, “6DOF” refers to three position dimensions (e.g., up/down; left/right; forward/back) and three axes of rotational motion (e.g., yaw; pitch; roll). In some such examples, the Hounsfield density of each point of reference may be substantially greater than a Hounsfield density of the body of the patient **620**, so as to easily distinguish the points of reference within the internal imagery. Additionally or alternatively, internal imaging target **628** may include an object having at least four points of reference (e.g., the same four points or an additional four points), wherein each point of reference defines a different geometric shape so as to define an

unambiguous 6DOF orientation of internal imaging target **628** in 3-D space. Additionally or alternatively, internal imaging target **628** may include an object having at least four points of reference (e.g., the same four points or an additional four points), wherein each pair of distance between any two points is unique to the other pairwise distances as to define an unambiguous 6DOF orientation of internal imaging target **628** in 3-D space.

(73) In some examples, internal imaging target **628** is affixed to a mounting system at a first position, and external imaging target **618** is affixed to the mounting system at a second position that is a predetermined distance from the first position, such that the relative alignment between the internal and external coordinate systems are known. For instance, the mounting system may be rigidly mounted to a bed on which patient **620** lies while the internal imaging data is collected. In some such examples (but not all such examples), the mounting system may include a robotic C-arm, wherein internal imaging target **628** includes a plurality of radiopaque markers affixed to the robotic C-arm, and wherein external imaging target **618** includes a geometric shape that has an unambiguously determinable orientation in 3-D space. Additionally or alternatively, one or more components of EM sensor system **650** may be affixed to the mounting system at a third position that is a predetermined distance from the first position and/or the second position.

(74) In some examples, a camera (e.g., external imaging device **608**) on HMD **108** detects external imaging target **618** and determines a location and orientation in space of external imaging target **618** relative to HMD **108**, and by extension, the perspective of the XR user **110**. Similarly, HMD **108** may receive internal imaging data from, e.g., an MRI imaging device or a CT scanning device and determine a location and orientation in space of internal imaging target **628** within the internal imaging data relative to HMD **108** and the perspective of the XR user **110**. HMD **108** may also use imaging targets **618**, **628** to align a determined pose of a virtual representation of medical tool **602** to external imaging data and internal imaging data.

(75) In some examples, HMD **108** determines the relative position and orientation of the EM field generator **622** using the camera **608** mounted on HMD **108**. The EM positioning system **650** and imaging target system **618** thus output a pair of relative coordinate transformations. When combined, the relative position and orientation between the perspective of HMD user **110** and the EM position trackers **604** can be calculated (e.g., by computing device **606**, which may be integrated within HMD **108**). In some examples herein, additional visual overlays can be provided to help guide a medical procedure involving medical tool **602** (e.g., a TIPS procedure) for patient **620**. In some instances, system **600** includes an internal imaging device **610** configured to capture internal image data depicting an internal anatomy of patient **620**. In one illustrative, non-limiting example, patient **620** may undergo a CT scan prior to a TIPS procedure. In some such examples, the anatomy around the liver of patient **620** can be segmented to create a digital, 3-D geometric model **654** of the patient's internal anatomy, which can be overlayed onto real-time imagery (e.g., from external imaging device **608**) during the TIPS procedure, in order to produce a simulated “transparent” view of the patient's body. In one particular example, computing device **606** may use internal imagery (e.g., CT scan imagery) from internal imaging device **610** to segment the blood vessels of the patient's liver, in order to help guide the TIPS needle and sheath **602** from, for example, the inferior vena cava (IVC) and hepatic veins into the portal vein.

(76) In some examples in which system **600** uses internal imaging data (e.g., from a CT scan) in an overlay during an AR-TIPS procedure, computing device **606** is further configured to determine the position and orientation (e.g., a relative alignment) of the digital, 3-D internal imaging data from the perspective of the user **110** of HMD **108**. As one such example, an external imaging target **628** may be placed outside of the body of patient **620**, and may be used by computing device **606** to perform this additional coordinate transformation or relative-alignment determination.

(77) In some cases, internal imaging target **628** may include one or more particular physical parameters so as to define, and readily indicate, an unambiguous pose of external imaging target **628**, both within internal imaging data (e.g., CT scans) and within external imaging data (e.g.,

visual imagery captured by camera **608**). For instance, the geometry of internal imaging target **628** may be designed such that its position and orientation can be determined unambiguously from internal imaging data (e.g. from Digital Imaging and Communications in Medicine (DICOM) volumetric-pixel (or “voxel”) computed tomography (CT) scan data). The physical composition of internal imaging target **628** may be selected such that the target **628** can be scanned with a CT scanner, or other similar internal imaging device **610**, with good contrast. Internal imaging target **628** may include individual components that each have a well-defined attenuation (e.g. a narrow range of Hounsfield units). A well-defined attenuation (e.g., a material having a Hounsfield unit value sufficiently greater than the surrounding tissue of the patient) can help computing device **606** to locate internal imaging target **628** within internal imaging data (e.g., 3-D voxel data) by filtering out all voxels outside of a particular range of attenuations.

(78) In some examples herein, internal imaging target **628** may be fixed in place relative to patient **620** and scanned, along with the patient **620**, by internal imaging device **610**. During anatomy segmentation, a position of the anatomy relative to internal imaging target **628** is known. During a subsequent TIPS procedure, the position of internal imaging target **628** may be registered to the external imaging device (camera) **608**, and/or the EM position-tracking system **650**. In some cases, the internal imaging target **628** is distinct from external imaging target **618**. In other cases, internal imaging target **628** includes an affixed EM position tracker. Finally, internal imaging target **628** may have mechanical features which can “interlock” with EM field generator **622** in only one configuration (e.g., one unambiguous pose). When EM field generator **622** and internal imaging target **628** are interlocked in this way, the position of each, relative to the other, is known. In some cases, EM field generator **622** may itself function as internal imaging target **628**. That is, EM field generator **622** may be scanned, by internal imaging device **610**, along with patient **620** such that their relative poses are known.

(79) In some examples, computing system **606** is configured to determine a pose of medical tool **602** relative to the internal anatomy of patient **620** based on a previous trajectory of medical tool **602**, as indicated by the signals indicative of the magnitude of the EM field received from EM sensor(s) **604**. For example, computing system **606** may be configured to determine and monitor a path of motion of EM sensor(s) **604** and compare a geometric shape (e.g., a curvature) of the path of motion to a geometric shape within the internal anatomy of the patient. In other words, computing system **606** may attempt to “match” the trajectory of EM sensor(s) **604** to a particular lumen, such as the interior of a vessel or vein of patient **620** (as depicted within internal imaging data), in order to “locate” EM sensor(s) **604** within the patient's body. In some such examples, computing system **606** may be configured to generate XR content **612**, such as a simulation of medical tool **602** at a particular location and/or trajectory within the internal anatomy of patient **620**, overlaid upon a corresponding portion of the exterior of the body of patient **620** depicted within the second (external) image data. In some examples, the computing system **606** may improve the registration between the real anatomy of a patient and the analogous virtual anatomy **654** using an algorithm which minimizes the distances between the interior space of the virtual anatomy **654** and the path of motion of the EM sensor(s) **604** over time.

(80) For instance, computing device **606** may use the path of motion of the EM sensors **604**, constrained by, for example, known positions of vessel walls of the patient's vasculature, to refine the alignment between real and virtual anatomy of patient **620**. In some such examples, computing device **606** may be configured to substitute this path-of-motion tracking for determining a relative alignment based on an internal imaging target **628**. As one example, computing device **606** may filter out “anomalous” or outside-threshold datapoints recorded by EM sensors **604**, such as datapoints with random artifacts, datapoints indicating that a rate of motion of the sensor was outside an expected range, datapoints indicating that a distance between two sensors was outside an expected range, or the like. Computing device **606** may then filter out additional datapoints indicating a position of sensor **604** that is outside an expected vein or vessel geometry, reduce or

minimize a positive-signed distance of the motion path points outside of an expected vein geometry (e.g., only add up positive distances outside veins, ignore negative distances inside veins), and/or reduce or minimize the distance between the datapoints and a central longitudinal axis of an expected vein geometry, in order to determine a relative alignment between EM sensors **604** and the internal anatomy of patient **620**.

(81) FIGS. 7A-7C illustrate various examples of multi-functional fiducial markers for aligning two or more distinct types of datasets, as described above. For instance, FIG. 7A is a conceptual diagram illustrating an example fiducial marker **700A**, which may be a 2-D marker (e.g., an image printed on a surface) or a physical, 3-D object. Fiducial marker **700A** includes at least three elongated arms **702A-702C** (collectively, “arms **702**,” or in the alternative, “arm **702**”) extending radially outward from a common center point **704**. As shown in FIG. 7A, each arm **702** extends a different distance away from center point **704** (e.g., defines a different length), and is oriented at a different angle **706** with respect to the other two arms. In this way, fiducial marker **700A** defines a single, unambiguously determinable orientation (e.g., having six-degrees-of-freedom (6DOF)), in 3-D space. Accordingly, fiducial marker **700A** may be used at least as an external imaging target **618**.

(82) In examples in which fiducial marker **700A** is formed from a radiopaque material, fiducial marker **700A** may additionally be used as an internal imaging target **628**, thereby providing “dual” reference-point functionality. In examples in which fiducial marker **700A** has a fixed, known, or readily determinable pose relative to medical tool **602**, and/or wherein any of arms **702** include embedded EM sensors **604**, fiducial marker **700A** may additionally be used as an EM reference point for EM system **650**, thereby providing “treble” reference-point functionality.

(83) As further shown in FIG. 7A, arms **702A-702C** may be formed to have substantially different shapes (e.g., rectangular, rounded, and triangular shapes, respectively). Such configurations which may further help image-processing software (e.g., pattern recognition) of computing device **606** to more quickly and/or more easily distinguish the orientation of fiducial marker **700A**, particularly when depicted at an oblique orientation within a 2-D dataset (e.g., internal or external image data) that either obscures one of the arms **702** or distorts the “apparent” lengths of arms **702** and/or the angles **706** therebetween.

(84) FIG. 7B is a conceptual diagram illustrating another fiducial marker **700B** that includes at least four points of reference **708**. Each point of reference defines a different geometric shape such the four points of reference **708** define an unambiguously determinable six-degrees-of-freedom (6DOF) orientation of the internal imaging target in three-dimensional (3-D) space when captured in external imaging data. In examples in which the points of reference **708** are at least partially radiopaque, fiducial marker **700B** may constitute a dual external-and-internal imaging target **618/628**.

(85) Additionally or alternatively, at least two of reference points **708** (e.g., points **708B**, **708D**) of fiducial marker **700B** define a separation distance **710A** that is unique from at least five other separation distances **710B-710F** (e.g., between all other pairwise distances between any other pair of reference points **708**), such the at least four points of reference **708** define an unambiguously determinable six-degrees-of-freedom (6DOF) orientation of the internal imaging target in three-dimensional (3-D) space.

(86) FIG. 7C is a conceptual diagram illustrating an example mounting system for various fiducial markers such that the relative poses (e.g., distances and orientations) between the markers are fixed and known, and accordingly, that the respective datasets may be easily aligned. In the example shown in 7C, the mounting system is depicted as a robotic C-arm **712**, such as for an X-ray machine. In such examples, the X-ray machine may constitute the internal imaging device **610**, configured to capture internal imaging data **714** depicting an internal anatomy **716** of a patient **620** (FIG. 6A).

(87) As shown in FIG. 7C, the fiducial-marker-mounting system includes an internal imaging target

628 in the form of a plurality of radiopaque markers **718** affixed to the robotic C-arm **712** such that the position and orientation of the imaging target **628** is known. In some such examples, internal imaging target **628** comprises an object having at least four points of reference (e.g., each radiopaque marker **718**), each point of reference comprising a different degree of radiopacity, or “Hounsfield” density, such that the four points of reference **718** collectively define an unambiguously determinable six-degrees-of-freedom (6DOF) orientation of the internal imaging target **628**.

(88) Additionally or alternatively, EM sensor system **650** (e.g., including at least EM generator **622**) is rigidly coupled to the mounting system **712** at a position that is a known or predetermined distance **720** from the internal imaging target **628**. Additionally or alternatively, an external imaging target may be rigidly coupled to the mounting system at a known or predetermined distance from either of internal imaging target **628** or EM system **650**. In the example depicted in FIG. 7C, the internal imaging target **628** may likewise function as an external imaging target, in that the radiopaque markers **718** are in an asymmetrical arrangement defining an orientation is unambiguously determinable within external imaging data.

(89) FIG. 8A is a flowchart illustrating an example technique for determining a “rough” or approximate alignment (e.g., respective relative positions and orientations) of an array of fiducial markers **928** having known sizes and densities. Fiducial markers **928** are examples of internal imaging target(s) **628** of FIGS. 6A and 6B. The alignment techniques of FIG. 8A may be performed, for example, by computing device **606** of FIG. 6A, and are described for purposes of illustration with respect to the two-dimensional (2-D) pixel arrays depicted in FIGS. 9A-9D. However, the techniques of FIG. 8A apply similarly to three-dimensional (3-D) voxel arrays, mutatis mutandis (e.g., by substituting “area” with “volume,” etc.).

(90) After a patient standard has been imaged (e.g., CT-scanned, MRI-scanned, etc.), a computing system **606** (FIG. 6) loads the resulting internal imaging data **902A** (FIG. 9A), such as Digital Imaging and COmmunications in Medicine (DICOM) data (**802**). The internal imaging data **902A** includes a 2-D array of pixels (or a 3-D array of voxels, as appropriate), some of which represent the scanned internal imaging targets **928**, and others of which represent random noise in the internal imaging data file.

(91) Computing system **606** applies a density filter to the internal image data **902A**. For instance, the relative color (e.g., relative darkness) of each pixel corresponds to the relative density of the respective object imaged at that pixel's location. Accordingly, computing system **606** scans through each pixel of internal imagery **902A** and compares the color of that pixel to a range of “permissible” colors corresponding to the known densities of the fiducial markers **928** (e.g., measured in Hounsfield units). In other words, computing system **606** assigns to each pixel a Boolean value (“YES” or “NO”) indicating whether the color of the pixel falls within the expected density range for the fiducial markers **928** (e.g., shown in white in FIG. 9B), or whether the color of the pixel does not fall within the expected density range for the fiducial markers **928** (e.g., shown in black in FIG. 9B). Computing system **606** then outputs the resulting Boolean array **902B** shown in FIG. 9B. In this way, computing system **606** effectively eliminates all pixels that fall outside the expected color/density range (shown in black in FIG. 9B), and assimilates all pixels that fall within the expected color/density range to a common color (e.g., white) (**804**).

(92) Computing system **606** then scans the Boolean array **902B** and identifies all “islands,” or groups **904** of adjacent white pixels that are completely surrounded by black pixels (or vice versa, as appropriate). In some instances, such as when the Boolean array **902B** includes a large number of single-pixel islands **904**, subsequent image-processing by computing system **606** may become very slow. In some such instances, computing system **606** may be optionally configured to eliminate “exceptionally” small pixel islands **904**, or in other words, to “de-speckle” the Boolean array **902B**. For instance, computing system **606** may be configured to first apply an “erode” filter to reduce the size of all pixel islands by a predetermined proportion, thereby “shrinking” all of the

below-threshold-size pixel islands out of existence, and then to apply a “dilate” filter to re-expand all remaining pixel islands **904** back to their original size (**808**).

(93) After applying the density-based pixel filter, computing system **606** applies a size-based filter. For instance, computing system **606** may measure the area (or volume, as appropriate) of each pixel island **904** and determines whether the island **904** falls within an expected size range associated with the known sizes of the fiducial markers **928**. As shown in FIG. **9C**, computing system **606** then “eliminates” all pixel islands **904** of pixels that fall outside the expected size range, e.g., that are either too large or too small to be indicative of the position of a fiducial marker **928** (**810**).

(94) At this stage, the only remaining pixel islands **904** should indicate the positions of the fiducial markers **928** (e.g., four pixel islands **904**, in the example shown in FIG. **9C**). As shown in FIG. **9D**, computing system **606** may then determine the approximate center **906** of each pixel island (**812**), and then measure the pairwise distance **908** (e.g., distances **710** of FIG. **7B**) between the centers **906** of the remaining pixel islands (six pairwise distances **908**, in the example shown in FIG. **9D**), in order to determine the approximate relative alignment between the fiducial markers **928** (**814**).

(95) FIG. **8B** is a flowchart illustrating a technique for determining a precise (e.g., sub-pixel-level) relative alignment for the array of fiducial markers **928** of FIG. **9A**. For instance, the precise-alignment techniques of FIG. **8B** may be performed by computing system **606** of FIG. **6A** after one or more rough or approximate alignment techniques (e.g., the techniques of FIG. **8A**) have been performed for the array of fiducial markers **928**.

(96) Computing system **606** generates a first function or algorithm configured to receive (e.g., as user input) a set of input parameters and create, based on the parameters, a virtual template of the array of fiducial markers **928** (**816**). For instance, the function may be configured to receive three “position” input parameters (x, y, z) and three “orientation” input parameters (yaw, pitch, roll). The subsequently generated virtual template may include virtual representations of the fiducial markers **928**, each virtual marker having a radiodensity equal to the average value of the known radiodensities of the physical fiducial markers **928**.

(97) Computing system **606** applies a second function or algorithm configured to evaluate (e.g., determine and indicate) how well the virtual template, at a particular position and orientation, matches some input volume data (**818**). For instance, computing system **606** may apply a cross-correlation function or other similar function to the virtual template for the given position and orientation.

(98) Computing system **606** then crops the original volume data from the DICOM data **902A** (FIG. **9A**) to retain only the pixels (or voxels, as appropriate) representing the fiducial markers **928** (**820**). Using the cropped DICOM data, computing system **606** computes an initial “guess” of the virtual template parameters (**822**), e.g., based on the determined relative positions of the centers **906** of the pixel islands **904**, as described above with respect to FIG. **8A**.

(99) Computing system **606** then applies the fitness function to perform gradient descent on the template alignment (**824**) to precisely determine the relative alignment of the fiducial markers **928**, e.g., up to a resolution that is higher than the resolution of the pixels of the DICOM data **902A**, in some instances.

(100) FIG. **10** is a flowchart illustrating example techniques for generating extended-reality (XR) content for a medical procedure. The techniques of FIG. **10** are described with respect to XR system **600** of FIGS. **6A** and **6B** for purposes of illustration, but may be performed by any suitable XR system in other examples.

(101) A computing device **606**, such as processing circuitry of a head-mounted-display (HMD) device **108**, receives, from one or more electromagnetic (EM) sensors **604** of an EM sensor system **650**, signals indicative of a magnitude of an EM field **624** generated by an EM source **622** of the EM sensor system **650** (**1000**). For instance, the one or more EM sensors **604** may be disposed at a distal portion of a flexible shaft **614** of a medical tool **602** configured for insertion into a body of a

patient **620**. As one example application of the techniques of FIG. **10**, the medical tool **602** may include a needle for a Transjugular Intrahepatic Portosystemic Shunt (TIPS) procedure.

(102) In response to receiving the EM sensor signals, computing device **606** determines, based on the signals indicative of the magnitude of the EM field **624**, a pose of the one or more EM sensors **604** (**1002**). For instance, the computing device **606** may determine a position and orientation of the EM sensor **604** relative to a known reference point, such as to a base **616** of the medical tool **602**, to the EM generator **622**, or to another fiducial reference point.

(103) The computing device **606** adjusts a virtual three-dimensional (3-D) model representation **652** of the medical tool **602** based on the determined pose of the one or more EM sensors **604** (**1004**). For instance, the computing device may determine or estimate a position and orientation of a distal tip of the flexible, elongated shaft **614** of the medical tool relative to the base **616**, and by extension, subsequently determine a shape (e.g., a bending or curvature) of the shaft **614** therebetween.

(104) The computing device **606** receives, from an internal imaging device **610**, a set of “first” image data or “internal” image data depicting at least a portion of an internal anatomy of the body of a patient **620** (**1006**). For instance, the computing device **606** may load DICOM data, including X-ray data, MRI data, CT scan data, or any other data depicting a relevant internal anatomy of patient **620**. In some examples, but not all examples, the received internal image data may depict or capture an internal imaging target **628** or other fiducial marker, e.g., having one or more known physical parameters.

(105) The computing device **606** receives, from an external imaging device **608** (e.g., a camera of HMD **108**), a set of “second” image data or “external” image data depicting at least a portion of the exterior of the body of patient **620** (**1008**). In some examples, but not all examples, the received external image data may depict or capture an external imaging target **618** or other fiducial marker, e.g., having one or more known physical parameters.

(106) The computing device **606** determines, based on the EM sensor signals, the first image data (e.g., the internal imaging target **628**), and the second image data (e.g., the external imaging target **608**), a relative alignment between at least two of the three datasets (**1010**). For instance, computing system **606** may determine a pose (e.g., a relative position and orientation) of the one or more EM sensors **604**, a pose of the external imaging target **618**, and/or a pose of the internal imaging target **628**. Based on the determined relative alignment between these reference points, computing device **606** may determine a relative alignment between the respective datasets as a whole. In some examples herein, a “treble” fiducial marker may constitute a common reference point for aligning all three datasets.

(107) The computing device **606** then generates, based on the determined relative alignment(s) between the reference points and/or the datasets, extended reality (XR) content **612** that includes a visual representation **652** of the medical tool **602** located at the corresponding “real-world” position within a visual representation **654** of at least a portion of the internal anatomy of the patient **620** (**1012**). The computing device **606** then customizes and outputs the XR content **612** for display on the appropriate XR display device **108** to simulate a current pose of the medical tool relative to the body of the patient (**1014**). For instance, in examples in which HMD **108** includes a transparent display screen, computing device **606** “registers” and tracks the position of the physical body of patient **620** relative to the transparent display, and displays the XR content **612** on the transparent display at the appropriate real-world position relative to the patient **620**. In examples in which HMD **108** includes an opaque display screen, computing device **606** generates composite imagery of the XR content **612** overlaid onto 2-D imagery of the exterior of the patient **620**, and outputs the composite imagery for display on the screen.

(108) The techniques described in this disclosure may be implemented, at least in part, in hardware, software, firmware or any combination thereof. For example, various aspects of the described techniques may be implemented within one or more processors, including one or more

microprocessors, digital signal processors (DSPs), application specific integrated circuits (ASICs), field programmable gate arrays (FPGAs), or any other equivalent integrated or discrete logic circuitry, as well as any combinations of such components. The term “processor” or “processing circuitry” may generally refer to any of the foregoing logic circuitry, alone or in combination with other logic circuitry, or any other equivalent circuitry. A control unit comprising hardware may also perform one or more of the techniques of this disclosure.

(109) Such hardware, software, and firmware may be implemented within the same device or within separate devices to support the various operations and functions described in this disclosure. In addition, any of the described units, modules or components may be implemented together or separately as discrete but interoperable logic devices. Depiction of different features as modules or units is intended to highlight different functional aspects and does not necessarily imply that such modules or units must be realized by separate hardware or software components. Rather, functionality associated with one or more modules or units may be performed by separate hardware or software components, or integrated within common or separate hardware or software components.

(110) The techniques described in this disclosure may also be embodied or encoded in a computer-readable medium, such as a computer-readable storage medium, containing instructions. Instructions embedded or encoded in a computer-readable storage medium may cause a programmable processor, or other processor, to perform the method, e.g., when the instructions are executed. Computer readable storage media may include random access memory (RAM), read only memory (ROM), programmable read only memory (PROM), erasable programmable read only memory (EPROM), electronically erasable programmable read only memory (EEPROM), flash memory, a hard disk, a CD-ROM, a floppy disk, a cassette, magnetic media, optical media, or other computer readable media.

(111) Various examples have been described. These and other examples are within the scope of the following claims.

Claims

1. A method comprising: receiving, by processing circuitry and from one or more electromagnetic (EM) sensors of an EM sensor system, signals indicative of a magnitude of an EM field generated by an EM source of the EM sensor system, wherein the one or more EM sensors are disposed at a distal portion of a flexible shaft of a medical tool configured for insertion into a body of a patient; determining, by the processing circuitry and based on the signals indicative of the magnitude of the EM field, a pose of the one or more EM sensors; adjusting, by the processing circuitry and based on the determined pose of the one or more EM sensors, a three-dimensional (3-D) model representation of the medical tool; receiving, by the processing circuitry, first image data depicting at least a portion of an internal anatomy of the body of the patient and an internal imaging target; receiving, by the processing circuitry, second image data depicting an external imaging target; determining, by the processing circuitry and based on the signals indicative of the magnitude of the EM field, the first image data, and the second image data, a relative alignment between the pose of the one or more EM sensors, a pose of the external imaging target, and a pose of the internal imaging target; generating, by the processing circuitry and based on the determined relative alignment, extended reality (XR) content comprising a visual representation of the medical tool within the at least a portion of the internal anatomy of the body; and outputting, by the processing circuitry, the XR content for display on an XR device to simulate a current pose of the medical tool relative to the body of the patient.

2. The method of claim 1, wherein determining the relative alignment comprises determining, by the processing circuitry, a first relative alignment between the determined pose of the one or more sensors and a pose of the internal imaging target, and wherein the method further comprises

generating, by the processing circuitry and based on the adjusted 3-D model representation of the medical tool and based on the first relative alignment between the determined pose of the one or more EM sensors and the pose of the internal imaging target, the visual representation of the medical tool within the at least a portion of the internal anatomy of the body.

3. The method of claim 2, wherein determining the pose of the one or more EM sensors comprises determining the pose of the one or more EM sensors with respect to an EM coordinate system, wherein the method further comprises determining the pose of the internal imaging target with respect to an internal imaging coordinate system, and wherein determining the first relative alignment between the pose of the one or more EM sensors and the pose of the internal imaging target comprises determining, based on a predefined distance between the EM sensor system and the internal imaging target, at least one of: the pose of the one or more EM sensors with respect to the internal imaging coordinate system; or the pose of the internal imaging target with respect to the EM coordinate system.

4. The method of claim 2, wherein determining the relative alignment further comprises determining, by the processing circuitry, a second relative alignment between the pose of the external imaging target and the pose of the internal imaging target.

5. The method of claim 4, wherein the method further comprises: determining the pose of the internal imaging target with respect to an internal imaging coordinate system; and determining the pose of the external imaging target with respect to an external imaging coordinate system, wherein determining the second relative alignment between the pose of the external imaging target and the pose of the internal imaging target comprises determining, based on a predefined distance between the internal imaging target and the external imaging target, at least one of: the pose of the internal imaging target with respect to the external coordinate system; or the pose of the external imaging target with respect to the internal coordinate system.

6. The method of claim 1, wherein the XR device comprises at least one of: a head-mounted display (HMD); a projector; a transparent display; a display reflected by a partially reflective surface; or a stationary screen-mounted display.

7. The method of claim 1, further comprising positioning the interior imaging target and the exterior imaging target with respect to a rigid portion of the body of the patient, wherein determining the relative alignment comprises determining the relative alignment based at least in part on distances of the interior imaging target and the exterior imaging target from the rigid portion of the body of the patient.

8. The method of claim 7, wherein the rigid portion of the body of the patient comprises a bony anatomy of the patient, and wherein determining the relative alignment based at least in part on the distances of the interior imaging target and the exterior imaging target from the rigid portion of the body of the patient comprises determining, with the one or more EM sensors, the distances of the interior imaging target and the exterior imaging target from the bony anatomy of the patient.

9. The method of claim 1, wherein the internal imaging target comprises an asymmetrical, radiopaque marker.

10. The method of claim 1, wherein the method further comprises determining, by the processing circuitry and based on the pose of the one or more EM sensors and the relative alignment, a trajectory of the medical tool through the at least a portion of the internal anatomy, and wherein generating the XR content comprises generating a visual representation of the trajectory of the medical tool through the at least a portion of the internal anatomy.

11. The method of claim 1, wherein the internal image data comprises CT-scan image data, X-ray image data, fluoroscopy image data, or MRI image data.

12. The method of claim 1, wherein the medical tool comprises a transjugular intrahepatic portosystemic shunt (TIPS) set.

13. The method of claim 1, wherein the second image data further depicts at least a portion of an exterior of the body of the patient.

14. The method of claim 1, wherein the internal imaging target comprises a set of radiopaque markers in an asymmetrical arrangement.
15. The method of claim 1, wherein the internal imaging target comprises an object having at least four points of reference, at least one pair of points has a distance unique from at least five other distances such that the at least four points of reference define an unambiguously determinable six-degrees-of-freedom (6DOF) orientation of the internal imaging target in three-dimensional (3-D) space.
16. The method of claim 1, wherein the medical tool comprises a prostate artery embolization catheter.
17. The method of claim 1, wherein the medical tool comprises a microcatheter.
18. The method of claim 1, wherein the medical tool comprises a guidewire.
19. A system comprising processing circuitry configured to: receive, from one or more sensors of an EM sensor system, signals indicative of a magnitude of an EM field generated by an EM source of the EM sensor system, wherein the one or more EM sensors are disposed at a distal portion of a flexible shaft of a medical tool configured for insertion into a body of a patient; determine, based on the signals indicative of the magnitude of the EM field, a pose of the one or more EM sensors; adjust, based on the determined pose of the one or more EM sensors, a three-dimensional (3-D) model representation of the medical tool; receive first image data depicting at least a portion of an internal anatomy of the body of the patient and an internal imaging target; receive second image data depicting an external imaging target; determine, based on the signals indicative of the magnitude of the EM field, the first image data, and the second image data, a relative alignment between the pose of the one or more EM sensors, a pose of the external imaging target, and a pose of the internal imaging target; generate, based on the determined relative alignment, extended reality (XR) content comprising a visual representation of the medical tool within the at least a portion of the internal anatomy of the body; and output the XR content for display on an XR device to simulate a current pose of the medical tool relative to the body of the patient.
20. A system comprising: a medical tool configured for insertion into a body of a patient; an electromagnetic (EM) sensor system comprising: an EM source configured to generate an EM field; and one or more EM sensors configured to generate signals indicative of a magnitude of the EM field, wherein the one or more EM sensors are disposed at a distal portion of a flexible shaft of the medical tool; processing circuitry configured to: receive, from the one or more EM sensors, the signals indicative of the magnitude of the EM field; determine, based on the signals indicative of the magnitude of the EM field, a pose of the one or more EM sensors; adjust, based on the determined pose of the one or more EM sensors, a three-dimensional (3-D) model representation of the medical tool; receive first image data depicting at least a portion of an internal anatomy of the body of the patient and an internal imaging target; receive second image data depicting an external imaging target; determine, based on the signals indicative of the magnitude of the EM field, the first image data, and the second image data, a relative alignment between the pose of the one or more EM sensors, a pose of the external imaging target, and a pose of the internal imaging target; generate, based on the determined relative alignment, extended reality (XR) content comprising a visual representation of the medical tool within the at least a portion of the internal anatomy of the body; and output the XR content for display on an XR device; and the XR device, where in the XR device is configured to display the XR content to a user to simulate a current pose of the medical tool relative to the body of the patient.
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